

Response to Reviewers

JRESS-D-26-04700

Similarity-Calibrated Conformal prediction: data-free coverage guarantees for remaining-useful-life intervals under operating-regime transfer

To Professor Cepin and the Reviewers

We thank both reviewers for reports that were unusually specific and, in two places, correct about problems we had not seen. The revision is substantial: one comment invalidated the calibration scheme behind every number in Section 5, and we re-ran the entire experimental campaign rather than patch around it. All twelve comments are addressed below, and five changes are worth flagging first.

1. The calibration was not i.i.d., and every number has been recomputed. Reviewer 1’s comment 5 was correct. Scores stacked at three monitoring points per unit correlate at $\rho = 0.908$, a design effect of 2.82, so the published calibration set of 1200 scores carried the information of roughly 426 independent ones. The primary analysis is now unit-level, one score per unit. The headline survives: regime-blind conformal miscovers by 0.132 and SCC recovers coverage to within 0.011 of target, against 0.006 previously, and the change is what theory predicts, since losing a factor of three in stacking should inflate the finite-sample floor by $\sqrt{3} = 1.73$, giving $0.006 \times 1.73 = 0.010$ against 0.011 observed.

2. A second dataset, with the positive and the negative both reported. On NASA C-MAPSS FD002 and FD004, 509 run-to-failure engines across six operating regimes, the coverage gap falls from 0.590 to 0.060 and from 0.567 to 0.058. The a-priori certificate, however, does not transfer: neither physically motivated departure quantity predicts the residual gap, and the bound holds on 67 to 83% of held-out regime pairs against 100% on the testbed. We report this as the boundary of the contribution rather than omitting it.

3. The certificate claim is narrowed accordingly. The abstract and conclusions now claim the a-priori bound only where the governing degradation law identifies the departure quantity ψ , as ISO 281 does for rolling-contact fatigue and no comparable standard does for a turbofan gas path. The dimensionless calibration, which is the part a practitioner deploys, is claimed broadly and is now validated on external data.

4. The actionability limit is conceded, not argued. At $\eta = 2$ the interval back-off is 1.011, so the certified lower bound is non-positive on average. The word “graceful” has been removed from the manuscript and replaced by a stated operating envelope.

5. Two errors we found ourselves during the revision, both now fixed. The robustness table had been computed under the old stacked calibration, which would have left two calibration schemes side by side in one paper; it has been re-run unit-level. Its “held-out” bound-satisfaction rate was in fact computed in sample; it is now a genuine 60/40 split, as the caption always claimed.

Every number in the revised manuscript is reproduced by a committed script in the public repository, including the figures, which previously had no generator. The C-MAPSS files were verified against the NASA release by checksum: `train_FD002.txt` has MD5 `b6eaab2a6b589e5e41d43ca2f99e379b`.

Comment	Outcome	Where in the revised manuscript
R1.1 base-model complexity	addressed	Sec. 5.4, Table 2
R1.2 deployability	addressed	Sec. 5.8, Table 6
R1.3 terminology	accepted	Secs. 1, 5.8, 6, 7
R1.4 misspecification	addressed	Sec. 5.7, Table 5, Sec. 3.6
R1.5 temporal dependence	conceded	Sec. 3.4, Sec. 4.3, all of Sec. 5
R2.1 second dataset	addressed	Sec. 4.4, Sec. 5.5, Table 3, Fig. 2(b)
R2.2 ψ derivation	addressed	Appendix B, Tables B.1–B.2
R2.3 discrete-mode baseline	addressed	Sec. 5.6, Table 4, Fig. 2(a)
R2.4 actionability	conceded	Secs. 4.3, 5.2, Fig. 3
R2 minor 1 (m and the norm)	addressed	Secs. 3.3, 5.3, Appendix B
R2 minor 2 (bound tightness)	conceded	Secs. 5.3, 5.4, Table 1, Fig. 4
R2 minor 3 (L fit)	addressed	Sec. 3.5

Reviewer 1

Comment 1: dependence on base-model complexity

Although the authors claim that the proposed method is applicable to any pre-trained predictor as a “drop-in modification,” the case study only tests extrapolation using an extremely simple nominal rate model. If a highly non-linear machine learning model (such as Random Forest or a Neural Network) were adopted as the base predictor, interpolation and extrapolation errors across operating regimes could become complicated. Consequently, there is a concern that the proposed dimensionless scale $\sigma(\theta)$ alone might not fully absorb the shift in the score distribution. The authors should evaluate how the conservatism of the a-priori bound changes when varying the complexity of the base model.

Response. We ran the experiment with four predictors of increasing capacity, each trained on the source condition only and deployed on the target, and we answer both halves of the comment: what happens to coverage, and what happens to the conservatism of the bound.

Coverage. The concern is well founded, and the damage falls entirely on the unscaled baseline. The naive coverage gap at $\eta = 0$ worsens by a factor of 3.5, from 0.127 for the closed-form extrapolator to 0.449 for a random forest, because a learner fitted flexibly to one operating regime fails harder when deployed on another. SCC is flat at 0.007 to 0.014 across all four classes, and under departure the more flexible learners are the better ones, 0.046 for a neural network against 0.102 for the closed form at $\eta = 2$, because a model fitted in the dimensionless coordinates absorbs part of the residual departure.

Conservatism of the bound. We fitted the certificate separately for each predictor class over the same configurations. Validity is unaffected: the bound holds on 100% of held-out configurations for all four, with margins in the narrow band 0.30 to 0.36, so its conservatism does not grow with model capacity. Its mechanism, however, does weaken, and we state that rather than leave it for a reader to find. The departure model explains the score-distribution mismatch well only for the closed-form predictor (held-out $R^2 = 0.90$, against -2.8 to -0.02 for the three learned ones), so with a learned predictor the bound holds on conservatism rather than because δ tracks the mismatch: the residual is then dominated by model error rather than by physical departure. This is the same pattern the C-MAPSS certificate shows in the reply to Reviewer 2’s major comment 1, and a further reason the certificate claim has been narrowed in this revision.

We also state the caveat the result carries: the drop-in property holds provided the learned predictor is fitted in the dimensionless coordinates, since no conformal layer can repair a point predictor that does not transfer.

We note one further finding rather than hide it. On C-MAPSS the ordering of the naive column runs the other way, because the raw sensor levels there encode the operating regime itself, so a flexible learner can partly infer the regime and compensate. What is invariant across both datasets is that SCC is the better arm in every row.

Changes in the manuscript. Second and third paragraphs of Section 5.4, the latter reporting the certificate fitted by predictor class, and two new columns in Table 2.

Comment 2: deployability and the data environment the method needs

In the FEMTO dataset, the runtime diagnostic returns an “indeterminate” status and safely falls back, which the authors positively evaluate as an “honest applicability limit.” However, in the practical field of reliability engineering (PHM), if even a widely and commonly used benchmark dataset is rejected due to “data scarcity and variability,” the cases where this method can be truly considered “deployable” might be extremely limited. The authors

should quantitatively clarify the practical requirements (design guidelines) regarding what kind of data environment (e.g., minimum required number of units or signal-to-noise ratio) is necessary to yield a “holds” status.

Response. We have mapped the requirement on exactly the two axes the comment names: units per condition, and the unit-to-unit life scatter, which is the signal-to-noise ratio for this test, since the structural signal is the L10 life ratio between conditions and the noise is the spread within a condition.

The requirement is modest. Three units per condition suffice at a scatter of 0.10, six at 0.25 and ten at 0.50; only at a scatter of 0.80 does it rise to about forty units. The C-MAPSS fleet added in this revision, at roughly 260 units per regime, sits far inside the feasible region, so the method is not confined to unusually rich datasets.

The map also settles the status of the FEMTO result. With six bearings per condition and a sevenfold life spread, that dataset sits in the worst corner of the map, where *holds* is returned in 0.50 to 0.00 of trials, so its *indeterminate* verdict is predicted by the data environment rather than being a quirk of the dataset.

We also report a property of our own test that the map makes visible: since the reported verdict is the worst of three pairwise interval tests at a nominal 95% each, the achievable rate under exact similitude is about $0.95^3 \approx 0.86$, which is where the larger fleets settle. A multiplicity-corrected verdict would raise that ceiling and is the obvious refinement; we have left the test as submitted and stated the ceiling rather than redesign the diagnostic during revision.

Changes in the manuscript. New opening and Table 6 in Section 5.8, with the FEMTO paragraph now referring back to the map.

Comment 3: FEMTO is described as field data

In the Abstract, Conclusion, and Discussion, the FEMTO dataset is occasionally referred to as a “field benchmark” or “field data limit.” However, FEMTO (PRONOSTIA) consists of accelerated degradation data acquired in a laboratory testbed. Since validation using true field data—such as an “operating industrial process” which the authors truly imply—has not been conducted, the terminology should be consistently revised to terms like “experimental benchmark” to avoid misleading readers into believing that field validation has already been achieved.

Response. Agreed, and corrected throughout. FEMTO is now described as an accelerated laboratory test platform, and the word “field” is reserved for genuinely operational, in-service data, which this study does not have. The heading of the corresponding results subsection has been rewritten for the same reason. The second dataset added at Reviewer 2’s request is likewise identified as externally authored simulation rather than field data. The abstract no longer refers to FEMTO at all, having been restructured to carry the second dataset within the word limit; the diagnostic mechanism it exercises is described there, and the FEMTO result itself is reported in Section 5.8.

A committed script, `scripts/r13_terminology_sweep.py`, audits the manuscript source for any unqualified use of “field” and was run before every subsequent commit of this revision.

Changes in the manuscript. Abstract, Section 1 scope paragraph and contribution list, Section 5.8 title and body, Section 6 limitation (iii), Section 7.

Comment 4: what happens when the assumed physics is wrong

The proposed method assumes that accurate physical parameters (e.g., load, speed, temperature) and the governing equations are known a priori through the Buckingham Pi theorem. In actual complex industrial machinery, however, the governing equations for degradation (such as Arrhenius or Lundberg-Palmgren) are not always known. A sensitivity analysis or a theoretical discussion is required regarding the risk that the dimensionless correction could backfire (i.e., amplify the data shift) if the physical model is partially misspecified or if unknown environmental factors are present.

Response. The comment names two distinct risks and we now treat them separately.

Unknown environmental factors are the case the departure sweep already covers: the secondary Arrhenius channel of weight η is an unmodelled physical effect the scale does not know about, and Section 5.2 reports what it costs, together with the operating envelope requested by Reviewer 2’s major comment 4.

Partial misspecification of the governing law is new work. We perturb the assumed activation energy while holding $k(T_{\text{ref}})$ fixed, so only the between-condition behaviour of the scale is distorted, which is the part similitude depends on. When only the scale is wrong and the predictor is unaffected, the SCC gap stays at or below 0.05 out to $|\Delta E/E| = 31\%$, remains better than not scaling at all out to $+62\%$, and backfires, in the reviewer’s sense of being worse than the unscaled baseline, only at $+100\%$ (0.139 against 0.132). When a single wrong model drives both the predictor and the scale, which is the realistic case, the method is much more fragile and backfires already at -10% (0.072 against 0.061).

The failure is not silent. The runtime invariance check returns *holds* only at the correct scale ($g = 0.967$, interval $[0.92, 1.01]$) and *violated* at every misspecification tested, including the -10% point at which the coupled case first turns against the method ($g = 1.215$, $[1.16, 1.27]$). This is a capability the submitted manuscript did not claim: because the check tests the invariance of the dimensionless life, it is sensitive to misspecification of the scale itself as well as to a departure from similitude.

Changes in the manuscript. New Section 5.7 with Table 5; the dual role is now stated at the end of Section 3.6.

Comment 5: within-unit dependence in the calibration set

When applying conformal prediction to time-series or degradation data like RUL, strong temporal correlations (dependencies) exist in data continuously acquired from the same unit (history between time steps). Strictly speaking, this causes the assumptions of i.i.d. or exchangeability to waver even within the same operating regime. While the proposed method addresses cross-regime shifts among different units (spatial shifts) using a physical scale, the authors should explain the theoretical consistency and the impact of temporal dependencies within the same unit on the validity of the conformal prediction layer (e.g., whether block conformal approaches are necessary).

Response. The reviewer is right, and this was the most consequential comment in either report. The submitted calibration stacked one score per (unit, monitoring fraction), so each unit contributed three strongly dependent points. We measured the dependence rather than argue about it: pairwise correlations across monitoring fractions of 0.965, 0.883 and 0.877, mean $\rho = 0.908$, design effect $1 + (F - 1)\rho = 2.82$, so the published $n = 1200$ carried an effective n of about 426 against 400 units.

The primary analysis is now unit-level, one score per unit, and every number and figure in Section 5 has been recomputed on that basis. The finding is unchanged: naive conformal miscovers

by 0.132 while SCC recovers to 0.011. The change in the floor, from 0.006 to 0.011, is what theory predicts for the loss of a factor of three in stacking, $\sqrt{3} \times 0.006 = 0.010$.

We also report the stronger option the comment implies. Where a guarantee is wanted simultaneously at every monitoring point of a unit rather than at one sampled point, block conformal calibration on the per-unit maximum score is exchangeable across units by construction; it costs conservatism, a gap of 0.007 at $\eta = 0$ and 0.173 at $\eta = 2$ against 0.011 and 0.097 for single-point unit-level calibration. We report it as the option for multi-point monitoring and state explicitly that exchangeability is required across *units*, not across time steps within a unit.

Changes in the manuscript. New Section 3.4; the calibration scheme and the averaging convention are stated in Section 4.3; all of Section 5, Table 1 and Figures 2 to 4 recomputed unit-level.

Reviewer 2

Major comment 1: no dataset the authors did not design

The entire empirical demonstration of the coverage guarantee in Eq. (1) uses a single simulator (Section 4.2, the first-order Arrhenius catalyst-deactivation ODE) in which the similitude-departure parameter ψ is literally the knob η that the author sets. Sections 5.1 through 5.4 and Table 1 all report results from this one system. FEMTO (Section 5.5) only tests the diagnostic, which correctly declines to certify, so it never exercises the certificate itself. That means there is no case in the paper where the coverage bound is checked against a physical asset whose true departure was not authored by the same person running the estimator. You could validate the certificate itself, not just the diagnostic, on a second physically grounded dataset with enough pe condition replication to establish ground truth. For instance, a bearing fatigue dataset with multiple load and speed conditions and enough units per condition that the L10 comparison in Section 3.5 is well powered rather than indeterminate as it was for FEMTO's six bearings per condition.

Response. We accepted this and added a second dataset. On the choice of dataset: we looked first for the bearing fleet the comment suggests, and to our knowledge no public run-to-failure bearing dataset provides the replication the L10 comparison needs. The standard sets give roughly five to six units per operating condition, an order of magnitude short of the requirement mapped in Section 5.8 at the life scatter those datasets exhibit, and that shortfall is what produced FEMTO's *indeterminate* verdict in the first place, so a comparable bearing set would reproduce the same non-result. We therefore went to NASA C-MAPSS, FD002 (260 engines) and FD004 (249), six operating regimes with every engine visiting all of them, which gives roughly 260 units per regime, about $43\times$ FEMTO's replication, and which we did not author.

The dimensionless coordinate there is derived rather than fitted, from referred (corrected) gas-turbine parameters, and it is self-checking: the sea-level static regime returns $\theta = 1.0000$ and $\delta_p = 0.9999$, and the implied damage clock spans $3.5\times$ across the six regimes.

The calibration transfers. Regime-blind conformal loses a mean 0.590 of coverage across the 30 ordered regime pairs of FD002; referring the sensors to standard day is not enough on its own and still loses 0.550; calibrating in the gas-path deviation coordinate recovers coverage to within 0.060 of the 0.90 target, a $9.8\times$ reduction, and FD004 replicates it at 0.567 to 0.058. No target failure data is used. We also checked that this is not an artefact of a weak base predictor: with a random forest on raw sensors the naive gap is 0.128 rather than 0.590, and SCC still improves on it at 0.053.

The certificate does not transfer, and we report that. Neither departure candidate identifies the residual gap: the ambient referred distance correlates 0.289 with it on FD002 and 0.179 on FD004, and the healthy gas-path signature distance correlates -0.122 and -0.086 ; held out, both regressions explain nothing, with R^2 between -0.074 and 0.022 . The bound consequently holds on 67 to 83% of held-out regime pairs against 100% on the testbed. We therefore separate the two halves of the contribution in the abstract, in Section 5.5 and in the conclusions: the dimensionless calibration is claimed broadly and is now validated externally; the a-priori certificate is claimed only where the governing law identifies ψ .

Changes in the manuscript. New Section 4.4 (design and protocol), new Section 5.5 (results, both halves), Table 3, Figure 2(b), and the narrowed claim in the abstract and Section 7.

Major comment 2: ψ is never made concrete

Section 3.2 sketches how the dimensionless scale σ and departure ψ arise conceptually for two textbook cases, rolling-contact fatigue and Arrhenius kinetics, but ψ is never actually

derived for either one. Only the simulator's η stands in for ψ anywhere in the paper. A practitioner reading this paper and trying to apply SCC to their own asset has no worked example of identifying which Buckingham Π groups become the scale and which become the residual departure outside the one system the author designed. Please work through the Lundberg-Palmgren case in an appendix, derive the actual dimensionless groups from the rating-life relation, state explicitly what physical effect constitutes ψ for that model, and show at least one numerical instance of computing it from published bearing parameters.

Response. A new appendix works the split through for rolling-contact fatigue, the best-standardised degradation law in the reliability literature. Dimensional analysis on life, load, dynamic capacity and speed gives two groups, $\Pi_1 = tn$ and $\Pi_2 = P/C$, and Lundberg–Palmgren is precisely the relation between them, so the scale is $\sigma = 10^6(C/P)^p/(60n)$ in hours. ISO 281 then identifies what is left over: the modified life $L_{nm} = a_1 a_{ISO} L_{10}$ has $a_{ISO} = f(e_C C_u/P, \kappa)$, so for this law the standard itself enumerates ψ as the lubrication regime and the contamination group.

Evaluated on the FEMTO conditions, the appendix reproduces the $1.63\times$ characteristic-life ratio from first principles, and it yields an instruction an engineer would not guess from the rating-life formula: at a matched thermal state the residual departure is 0.091, while a 25°C spread in operating temperature raises it to 1.060, roughly $12\times$ larger. For bearings it is the thermal and lubrication state, not the load, that breaks similitude.

This also connects to major comment 1. Where the governing standard enumerates the residual groups, ψ is identifiable and the a-priori bound applies; no equivalent enumeration exists for a turbofan gas path, which is the most plausible reading of the C-MAPSS negative. Identifiability of ψ , rather than the bound itself, is the practical boundary of the method.

Changes in the manuscript. New Appendix B with Tables B.1 and B.2, cross-referenced from Sections 3.2, 5.5 and 5.8.

Major comment 3: comparison against discrete-mode conformal

Section 2 discusses Javanmardi and Hüllermeier [5] as the closest prior treatment of conformalized RUL, noting that their fix for varying operating conditions is to cluster into discrete modes. That comparison stays qualitative. Section 5's case study never runs the clustering approach as a third baseline alongside naïve and SCC conformal in Fig. 2. Since [5] is presented as the direct predecessor this paper improves on. Please add a discretized-temperature clustering baseline (treating the three testbed temperatures as [5]'s discrete modes) to Fig. 2, so the reader sees quantitatively what continuous physical scaling buys over discrete mode clustering rather than taking the qualitative contrast on faith.

Response. Added, with the most favourable reading given to the baseline: the three testbed temperatures are supplied to it as known modes. The comparison is honest in both directions.

When the target condition has its own run-to-failure history, discrete-mode calibration is near-oracle and, once the departure is large, better than SCC (0.009 against 0.045 at $\eta = 1$). We state plainly that in that situation the discrete remedy is the right tool and the physical scale is not needed.

When the target condition has no failure history, the discrete method has no calibration set for that mode, falls back to the nearest mode, and lands close to regime-blind pooling, 0.078 to 0.138 against 0.093 to 0.159 pooled, while SCC holds 0.009 to 0.040, a gap smaller by $3.5\times$ to $8.7\times$. That regime, a newly commissioned unit or a new operating point, is the one SCC exists for.

Changes in the manuscript. New Section 5.6 with Table 4, and a third series in Figure 2(a) showing that the nearest-mode fallback inherits the same directional failure as regime-blind

calibration.

Major comment 4: are the intervals actionable at large departures

Section 4.3 defines efficiency such that a diverging value “signals degenerate, uselessly wide intervals,” and Section 5.2 reports that at $\eta = 2$ the relative back-off reaches 1.01 (Fig. 3), meaning the certified interval half-width is roughly the size of the point RUL estimate itself. That same section describes this outcome as the gap and back-off degrading “gracefully.” In my understanding, those two statements are in tension: by the paper’s own definition of degeneracy, a back-off near 1.0 is close to the line the paper itself says marks an uninformative interval. Could you please state explicitly, using the paper’s own efficiency criterion, what back-off value should be read as the point where SCC’s output stops being actionable for a maintenance planner, and say plainly whether the $\eta = 2$ configuration in Fig. 3 is on the safe or the degenerate side of that line?

Response. The two statements were in tension, and the reviewer is right about which one has to go. We answer the two questions plainly.

Where actionability stops. Efficiency is now defined as the relative back-off $b = q_T / \overline{\text{RUL}}_T$, and because SCC returns a one-sided lower bound, the quantity a planner acts on is the certified fraction of predicted life, $1 - b$. Three bands are fixed in Section 4.3: $b < 0.5$ is actionable, since more than half the predicted life is certified; $0.5 \leq b < 1$ is degraded, a positive but shrinking lead time; and $b \geq 1$ is degenerate, because the certified lower bound is then non-positive on average, so the interval states only that failure has not yet occurred and carries no replacement schedule.

Which side $\eta = 2$ is on. The degenerate side. The measured back-off there is 1.011 and the certified fraction of predicted life is -0.011 . Coverage is still certified, with a gap of 0.097, and Section 5.2 says so, but the interval is not actionable. The envelope is now stated in full: SCC is actionable out to $\eta = 0.68$, degraded to $\eta = 1.97$, and degenerate beyond, with the band edges drawn on the figure. The word “graceful” has been removed from the manuscript entirely, in the abstract, Section 4.3, Section 6 and Section 7 as well as Section 5.2.

Changes in the manuscript. Efficiency and the bands defined in Section 4.3; the envelope stated in Section 5.2; bands shaded in Figure 3; “graceful” removed from the abstract, Section 4.3, Section 6 and Section 7.

Minor comment 1: the dimension of ψ and the choice of norm

Definition 1 (Appendix A) defines ψ as a vector in $\mathbb{R}^m$, but every case-study configuration varies a single scalar departure via η . Please state explicitly that the tested case is $m = 1$, and note whether the norm choice in Eq. (1) matters once $m > 1$ is actually exercised.

Response. Correct, and now stated. The dimension m is set by the physics rather than by the method, and the case exercised in this paper has $m = 1$, a scalar weight on a single unmodelled channel, so the norm reduces to an absolute value and its choice is immaterial. For $m > 1$ the norm must be fixed at the same time as ψ , since L is estimated in that norm and rescales with it. Appendix B gives the concrete two-dimensional case: carrying the contamination group alongside the lubrication regime makes ψ two-dimensional, and we say why the appendix uses the scalar $\ln \kappa$ instead.

Changes in the manuscript. Section 3.3, the scalar instance labelled in Section 5.3, and the $m = 2$ case in Appendix B.

Minor comment 2: the bound holds on 100% of configurations, which suggests looseness

Table 1 reports Bound % = 100 in every row. A bound that always holds could simply be loose everywhere rather than informative. You could report the average or worst-case gap between the bound and the measured coverage gap so a reader can judge tightness, not just validity.

Response. The suspicion is correct and we now report it. Across held-out configurations the mean certified bound is 0.384 against a mean measured gap of 0.046, a margin of 0.339 with a range of 0.268 to 0.552, so the certificate is conservative by roughly a factor of eight. We also name the two sources. The dominant one is the intercept rather than the physical slope, since at $\delta = 0$ the bound is $2a = 0.278$ against a measured gap of 0.011; that intercept is the finite-sample floor of the plug-in total-variation estimator and decays as $n^{-1/2}$ in the number of calibration units, from 0.424 at 50 units to 0.096 at 800, a log-log slope of -0.513 against the -0.500 predicted. The second is irreducible: the swap argument of Lemma 1 bounds the perturbation of the calibration set by twice the total-variation distance, and no amount of calibration data removes it. The certificate is presented as a worst-case planning bound rather than a sharp estimate.

Changes in the manuscript. Section 5.3; a margin column in Table 1; the slack shaded and the mean margin annotated in Figure 4.

Minor comment 3: how L is fitted

Section 3.4 describes regressing d_{TV} on $\|\Delta\psi\|$ “over source pairs,” which with three temperatures gives only three pairs, too few to fit a reliable slope. Section 5.3 clarifies that the fit actually uses multiple (pair, η) configurations. Move that clarification into Section 3.4 itself so the estimation procedure is not misread as relying on three data points.

Response. Clarified and moved to the method, where it belongs. The design points are configurations rather than conditions: each ordered pair of source conditions contributes one point per distinct departure magnitude the source data exhibits, so a fleet with three conditions supplies far more than three points. In the study reported in Section 5.3 this gives 30 (pair, departure) configurations, of which 60% are used for the fit and the remaining 40% are held out to validate the resulting bound.

Changes in the manuscript. Section 3.5.

One paper worth adding

Javanmardi and Hüllermeier [5] is already cited but only described qualitatively, not run as a baseline; see Major Comment 3 for how the authors can put it to use as an empirical comparison rather than a narrative contrast.

Response. It is now run as a baseline rather than described, as set out in the reply to major comment 3: Section 5.6, Table 4 and the third series in Figure 2(a).

Reproducibility

Every quantitative claim in the revised manuscript is reproduced by a committed script, and the figures are now generated by one as well. The C-MAPSS files were checksum-verified against the NASA Prognostics Data Repository release before use. We are grateful to both reviewers: the manuscript makes a narrower claim than it did, on considerably more evidence.